CATERPILLAR®

DEALER BEST PRACTICE SERIES

Measuring Oil Cleanliness

Application | Maintenance | Site Management | Component Rebuild | **Component Life Management** | MARC Management

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1.0 Introduction

Maximizing component life requires maintaining high levels of fluid cleanliness. **The ability to effectively and consistently measure debris in fluids is a key element in catching failures before they occur**. Laser particle counters are used to measure the amount of debris in oil. These units may be permanent installations in an oil analysis lab or portable units for field use.

Lab units are more repeatable and have less variability, but transportation to a lab and processing could take up to two days.

Portable units provide real-time information at the mine site but are more variable due to instrument quality and operator variability of how samples are processed. Both approaches have benefit and drawbacks.

2.0 Best Practice Description

Closely monitoring and tracking particle count data for each compartment is an effective way to manage fluid cleanliness and component performance. Particle counts before and after PM service help to identify the following;

2.1 Cleanliness at the Start of the PM Period

Compartment oil cleanliness after PM service, off-board filtration, etc. at the beginning of the PM interval.

2.2 Cleanliness at the End of the PM Period

Compartment oil cleanliness at the end of the PM interval prior to PM service. This indicates if the on-board filtration is capable of keeping the oil clean.

2.3 Break-In Period Completion

If end of PM readings exceed ISO 18/15, off-board filtration is recommended during PM to remove excess contamination. If oil cleanliness is ISO 18/15 or better, the filters are capable of maintaining fluid cleanliness and off-board filtration is not required.

2.4 Component Failure In Progress

After the break-in period is complete, particle counts will usually stabilize to +/- 1 ISO code range (due to measurement variability). If particle increase is more than the normal range of variability, a failure may be in progress. Failures almost always generate large amounts of debris that can usually be detected by particle count. SOS sample data should be used to verify if a failure is in progress.

3.0 Implementation Steps

3.1 Oil Analysis Lab

If an oil analysis lab is close, it is the lower cost option for particle count measurements. Some dealers and customers use portable counters for immediate results and oil analysis at PM or if the portable counter indicates a problem.

3.2 Portable Particle Counters

Portable counters are expensive, costing $10,000 or more per unit. However, they are rapidly gaining acceptance due to their convenience and real-time feedback. Particle counters are available with a variety of functions, including data recording.

3.3 Data Management

Looking at row after row of particle count numbers can be mind numbing. If particle count is to be used as a tool, it must be displayed in a manner which permits easy visual assessment of trends and abnormal increases. Particle count data for each compartment of each machine can be tracked using the attached software

3.4 Develop New Wear Material Trends

If replacing standard filtration with HE or UHE filtration, more debris will be captured and wear materials in the oil will increase at a much slower rate. New trends will be much lower than traditional level. New trends will need to be developed based on the performance with the improved filtration.

4.0 Benefits

1. Trending particle data for all particles provides an easy-to-use and convenient tool to monitor component health. If particle count raises sharply, an SOS sample can be used to determine the specific wear metal showing elevated levels.

2. Lower levels of contaminants extend the duration of failures in progress and allow more time to schedule repairs before a catastrophic failure occurs.

3. Portable counters allow real time multiple samples from all machine compartments, as well as new bulk fuel and bulk oils.

5.0 Resources Required

- Oil analysis lab or portable particle counter

- Access software program that allows input of particle count data, stores information in tables and provides output reports.
    - Developed and shared by Griff Jones (Unatrac).
    - Obtained through Jeff Wolffe, EAME Mining Rep. Wolffe_Jeffrey_S@cat.com (Available mid-July 2006)

- Advice on particle counter selection features and use available from Caterpillar Marketing & Product Support Division Contamination Control Group.
    - Contact Dave Baumann, Baumann_David_L@cat.com or Carmen Rose. Rose_Carmen_L@cat.com

6.0 Supporting Attachments

Component Life Management Master Document PDF file. (Click on Attachments within this document to view/open file)

7.0 Related Best Practices

0806-2.10-1000 -Managing Fluid Cleanliness

8.0 Acknowledgements

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